

THERMODYNAMIC ANALYSIS OF A HEAT SINK

GROUP 3

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INTRODUCTION

CONDUCTION

According to Fourier's law,

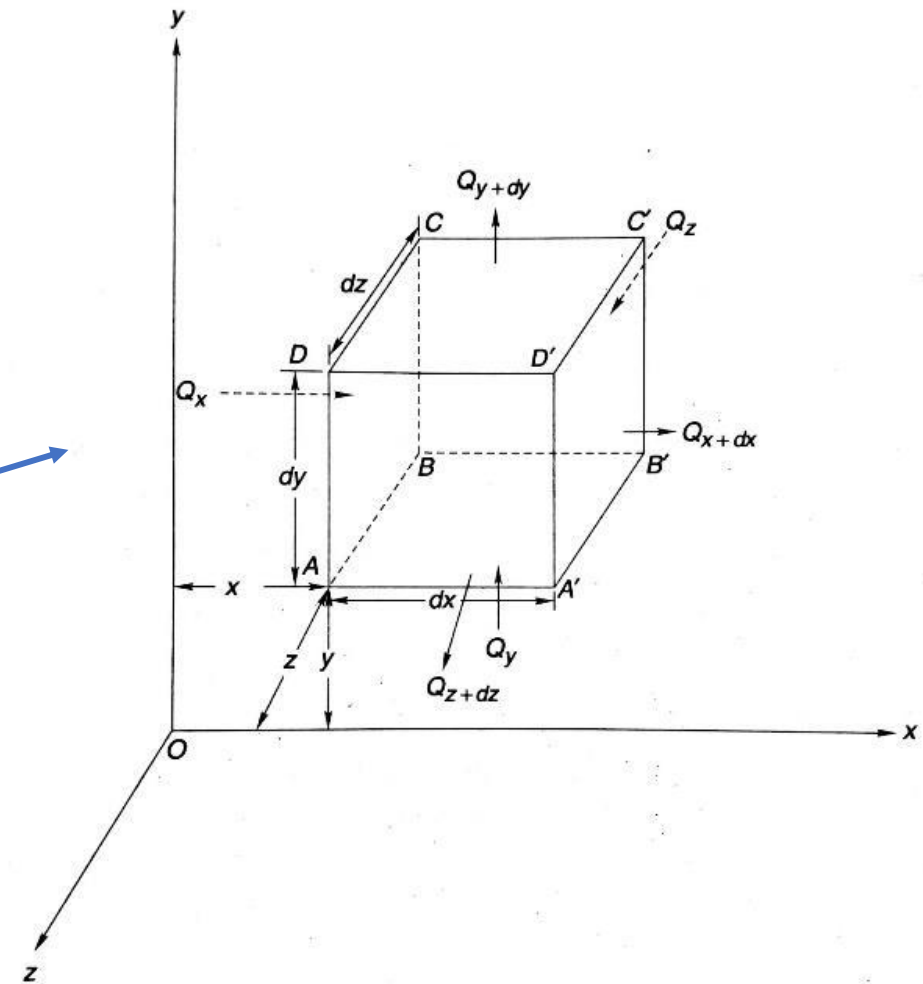
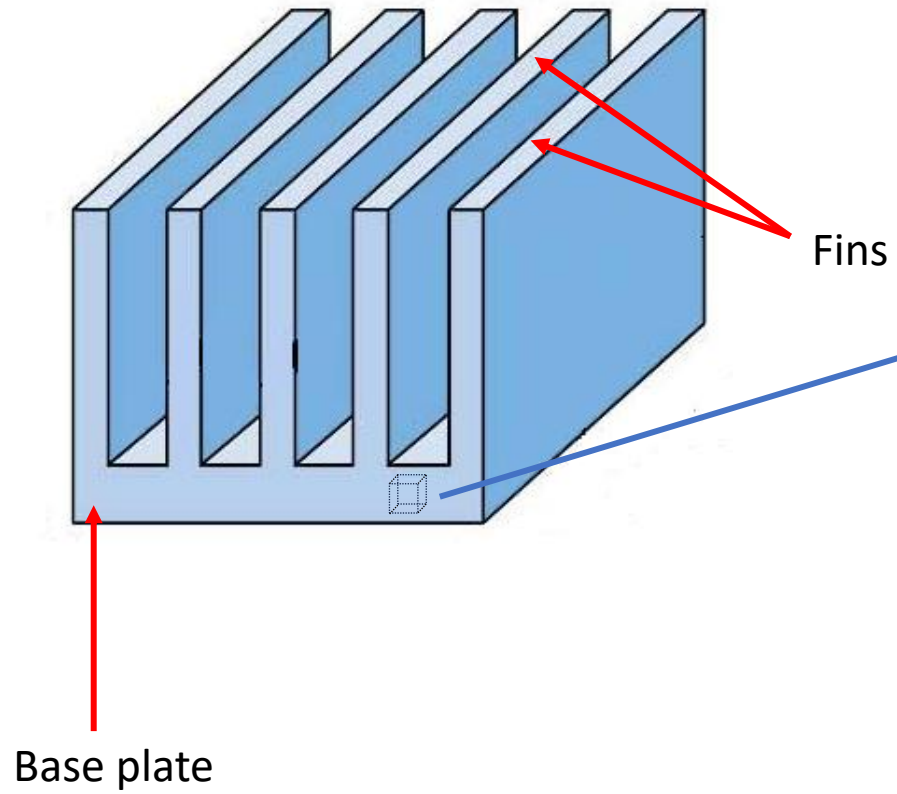
$$Q_x = -k_x A \frac{\partial T}{\partial x}$$

similarly,

$$Q_y = -k_y A \frac{\partial T}{\partial y}$$

$$Q_z = -k_z A \frac{\partial T}{\partial z}$$

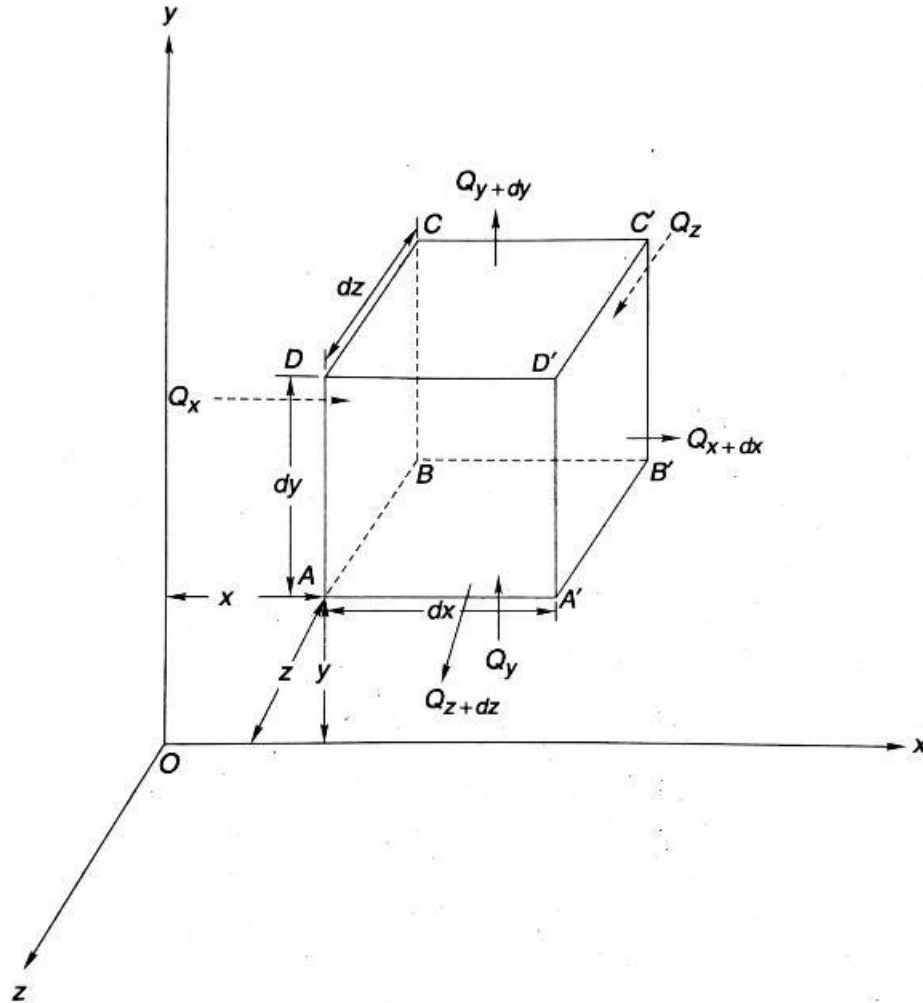
Where k_x, k_y, k_z are the thermal conductivities along the x-, y- and z-direction of the material, respectively.



The energy balance for this small element is obtained from the first law of thermodynamics as:

$$\begin{array}{ccccccc} \text{Net heat conducted} & & & & \text{Internal heat} & & \text{Increase internal} & & \text{Work done by} \\ \text{into element in x-} & & & & \text{generated per} & & \text{energy per unit} & & \text{element per unit} \\ \text{,y-,z-direction per} & + & & & \text{unit time} & = & \text{time} & + & \text{time} \\ \text{unit time} & & & & & & & & \\ \text{(I)} & & & & \text{(II)} & & \text{(III)} & & \text{(IV)} \end{array}$$

* The last term very small because the flow work done by solids due to temperature.



At x (at face ABCD), Let Q_x be the rate of heat conduction in x -direction and at $x + dx$ (at face A'B'C'D'), Q_{x+dx} be the rate of heat conduction.

According to Fourier's law, at x ,

$$Q_x = -k_x \frac{\partial T}{\partial x} dy dz$$

According to Taylor's series, at $x+dx$,

$$Q_{x+dx} = Q_x + \frac{\partial Q_x}{\partial x} \frac{dx}{1!} + \underbrace{\frac{\partial^2 Q_x}{\partial x^2} \frac{dx^2}{2!} + \dots}_{\text{Can neglected}}$$

The net rate of heat entering the element in x -direction,

$$Q_x - Q_{x+dx} = \frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) dx dy dz$$

The net rate of heat entering the element in x -, y -, z -direction per unit time

$$I = (Q_x - Q_{x+dx}) + (Q_y - Q_{y+dy}) + (Q_z - Q_{z+dz})$$

$$I = \left[\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) \right] dx dy dz$$

Let $\dot{q}$ be the rate of internal heat generation per unit volume, then the rate of energy generation in the element,

$$\dot{Q} = \dot{q} \, dx \, dy \, dz$$

The rate of change in internal energy,

$$= mc_p \frac{\partial T}{\partial t}$$

$$= \rho V c_p \frac{\partial T}{\partial t}$$

$$= \rho \, dx \, dy \, dz (c_p) \frac{\partial T}{\partial t}$$

$$\dot{Q} = \left(\rho c_p \frac{\partial T}{\partial t} \right) dx \, dy \, dz$$

where,

ρ - the density

c_p - specific heat

V - the volume

of the material of the element

Then, the general three-dimensional equation for heat conduction,

$$\underbrace{\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right)}_{\text{I}} + \underbrace{\dot{q}}_{\text{II}} = \underbrace{\rho c_p \frac{\partial T}{\partial t}}_{\text{III}}$$

For isotropic material, $k_x = k_y = k_z = k = \text{constant}$

The general three-dimensional heat conduction equation becomes

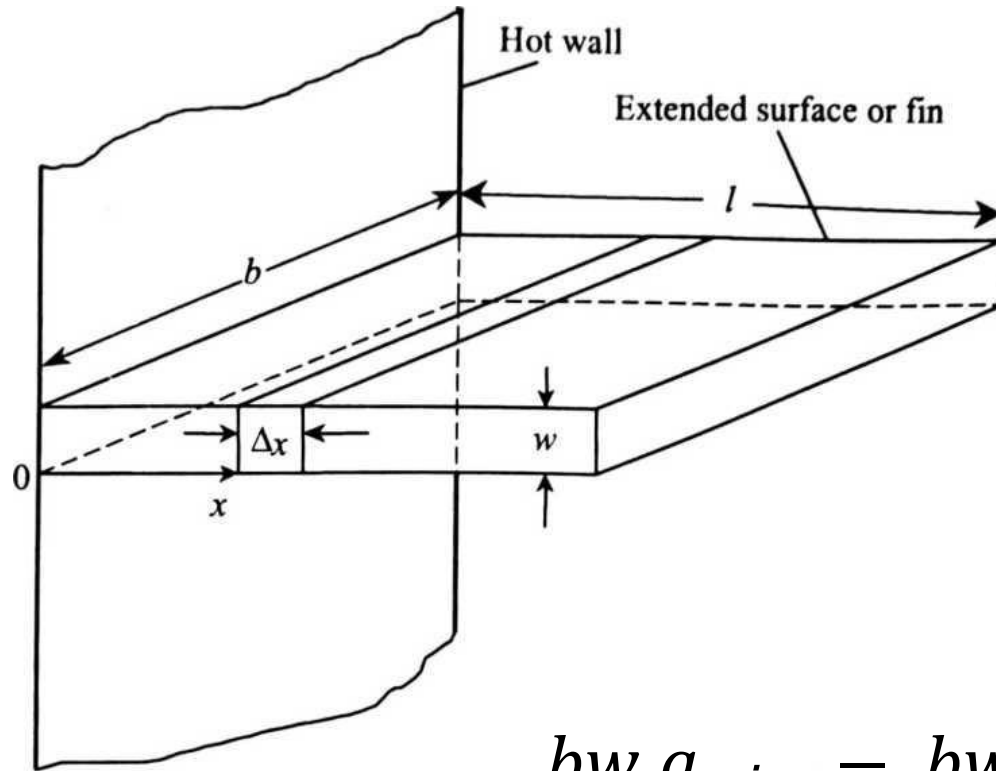
$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{\rho c_p}{k} \frac{\partial T}{\partial t} = \frac{\rho c_p}{k} \frac{\partial T}{\partial t}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{\rho c_p}{k} \frac{\partial T}{\partial t} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

The **thermal diffusivity**,

$$\alpha = \frac{k}{\rho c_p}$$

CONVECTION



Total Heat Input = Total Heat Output

$$bw q_{x \text{ in}} - bw q_{x \text{ out}} - 2(b + w)\Delta x h(T - T_0) = 0$$

Temperature at a point

$$T - T_0 = (T_w - T_0) e^{-mx}$$

T_w = Wall temperature

T_0 = Ambient Temperature

$$m = \sqrt{\frac{2(b+w)h}{kbw}}$$

Rate of Heat Transfer

$$Q = (T_w - T_0) \sqrt{2bw(b+w)hk}$$

h = Heat Transfer Coefficient

k = Thermal Conductivity

b = breadth

W = Thickness

Important thermodynamic correlations

(i) Nusselt Number - Nu

$$\text{Nusselt number} = \frac{\text{Wall temperature gradient}}{\text{Temperature gradient across the fluid in the pip}}$$

$$\text{Nu} = \frac{hl}{k} \quad \longrightarrow \quad \overline{\text{Nu}} = \frac{\bar{h}l}{k}$$

h - heat transfer coefficient

k - Thermal conductivity

l - length of the pin

$\overline{\text{Nu}}$ = Average Nusselt Number

$\bar{h}$ = heat transfer coefficient

(ii) Reynolds Number - Re

$$\text{Reynolds number} = \frac{\text{Inertial force}}{\text{Viscous force}}$$

$$\text{Re} = \frac{IV\rho}{\mu}$$

V - Average fluid velocity

ρ - Density

μ - Dynamic viscosity

(iii) Grashof Number - Gr

$$\text{Grashof number} = \frac{\text{Buoyancy force}}{\text{Viscous force}}$$

$$\text{Gr} = \frac{g\beta(T_s - T_o)L^3}{\nu^2} \quad \xrightarrow{\mu = \nu\rho} \quad Gr = \frac{g\beta\rho^2(T_s - T_o)L^3}{\mu^2}$$

β - Thermal Expansion Coefficient
 g - Acceleration due to Earth's gravity
 ν - Kinematic Viscosity
 μ - Dynamic Viscosity
 T_s - Wall Temperature
 T_o - Bulk Temperature
 L^3 - Length of the pin
 ρ - Density

$$\rho_i = \rho_f(1 + \beta dT) \quad \xrightarrow{\quad} \quad \beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P$$

ρ_i - Initial density
 ρ_f - Final density
 T - Temperature

(iv) Prandtl Number - Pr

$$\text{Prandtl number} = \frac{\text{Momentum Diffusivity}}{\text{Thermal Diffusivity}}$$

$$\mathbf{Pr} = \frac{\nu}{\alpha} = \frac{C_P \cdot \mu}{k}$$

C_P - Specific Heat Capacity

ν – Momentum Diffusivity

α – Thermal Diffusivity

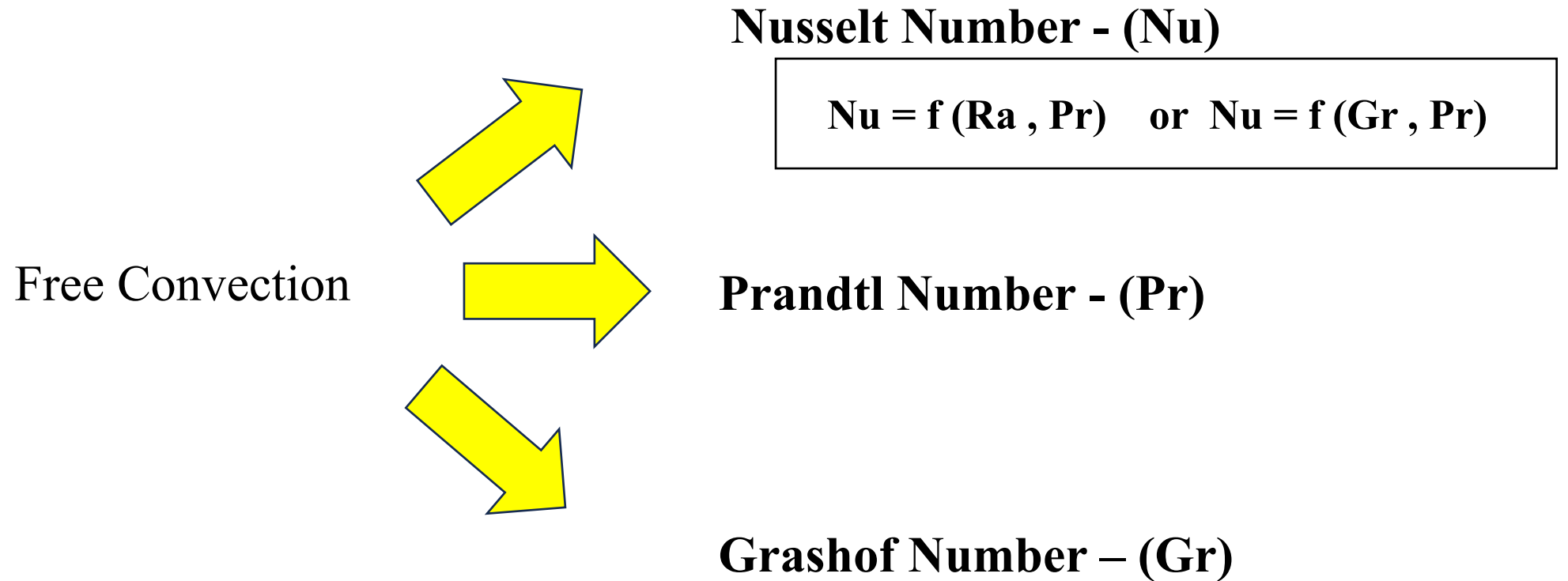
k - Thermal conductivity

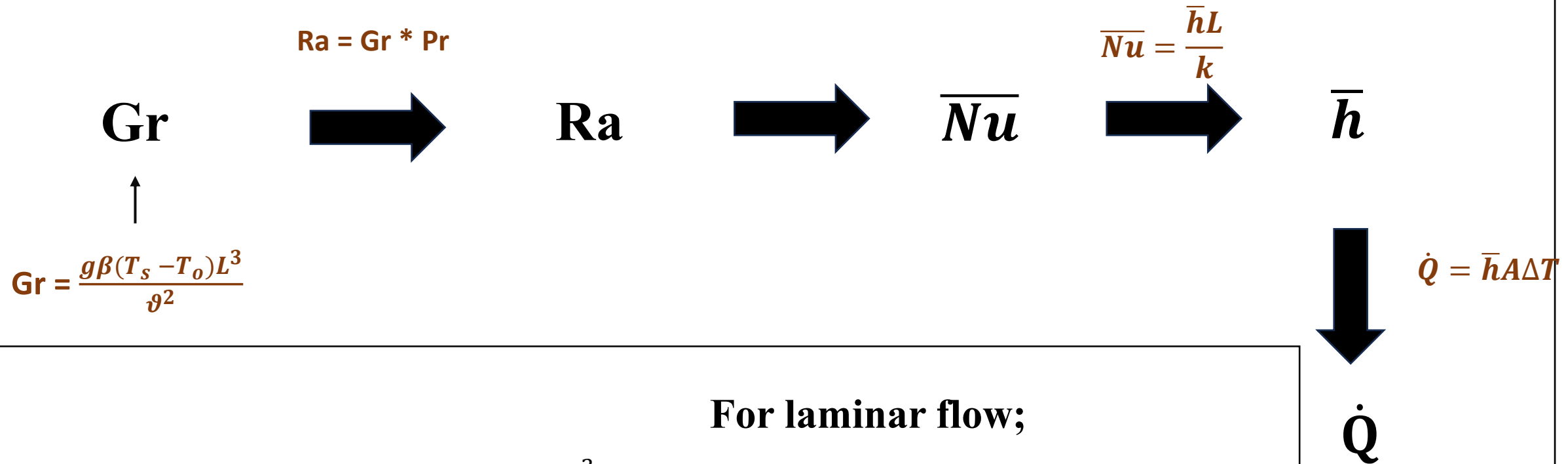
(v) Rayleigh Number - Ra

Rayleigh Number = (Grashof Number) x (Prandtl Number)

$$\mathbf{Ra = Gr * Pr}$$

Free Convection





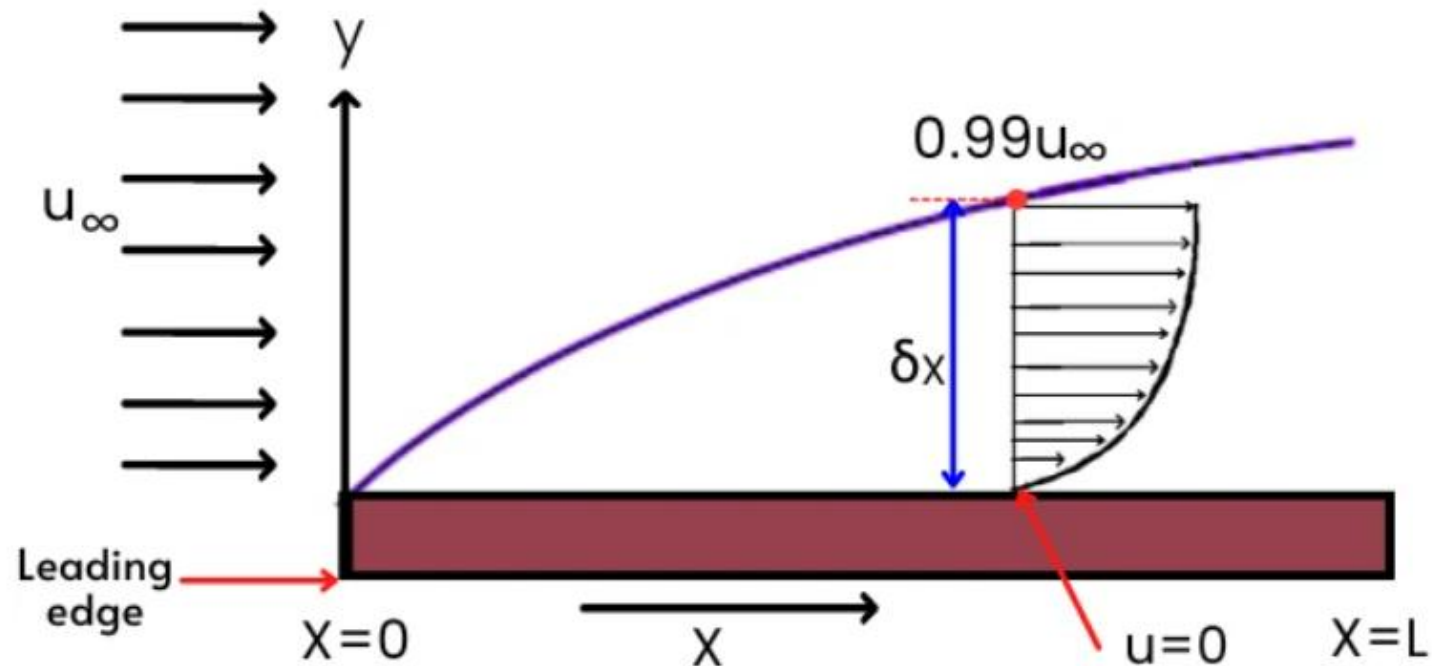
For laminar flow;

$$\overline{Nu} = \frac{\bar{h}L}{k} = \left\{ 0.825 + \frac{0.387(Ra)^{1/6}}{\left[1 + \left(\frac{0.492}{Pr} \right)^{9/16} \right]^{8/27}} \right\}^2 \Rightarrow \overline{Nu} = \frac{\bar{h}L}{k} = \left\{ 0.68 + \frac{0.67(Ra)^{1/4}}{\left[1 + \left(\frac{0.492}{Pr} \right)^{9/16} \right]^{4/9}} \right\}$$

Boundary layer

Definition

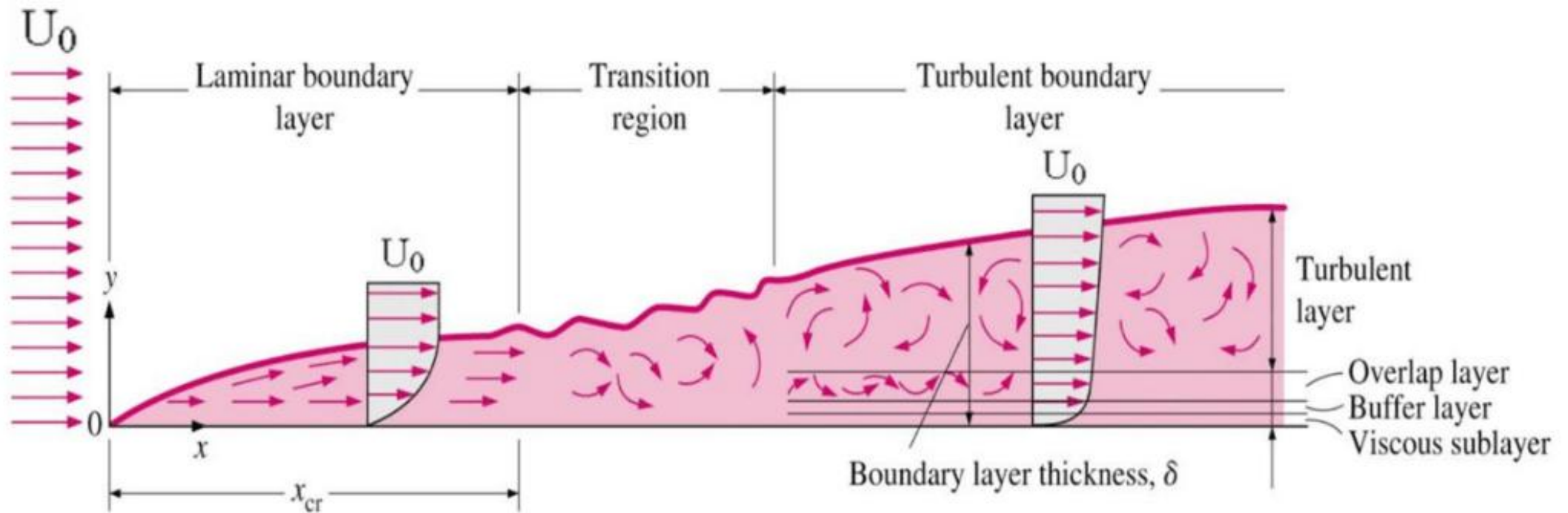
it represents the distance from the surface to the point where the flow velocity is approximately 99% of the velocity of the fluid far away from the surface. Definition



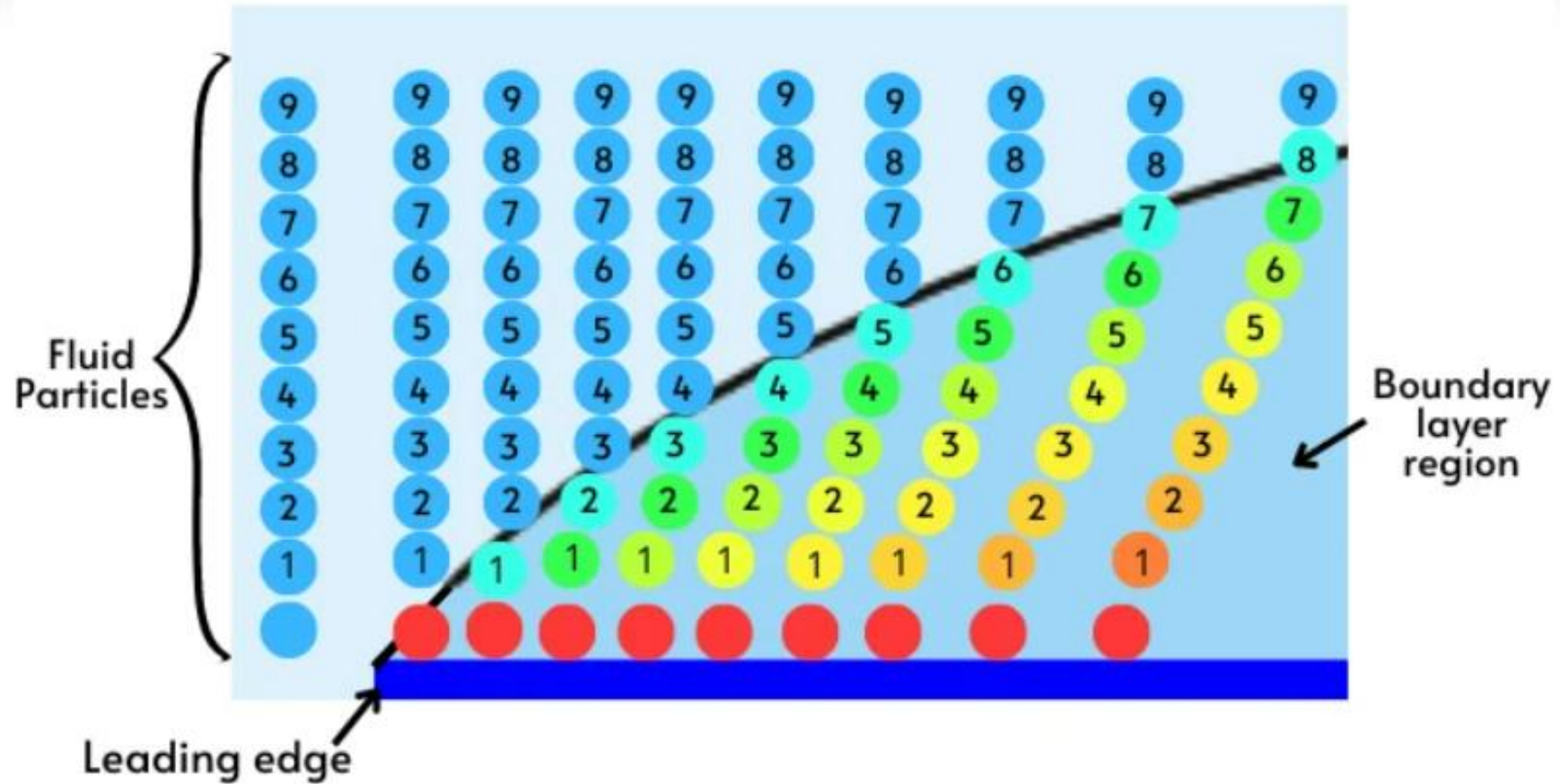
variations in boundary layer thickness and the different regions

If $Re_x < Re_{cr}$ then the flow is laminar.

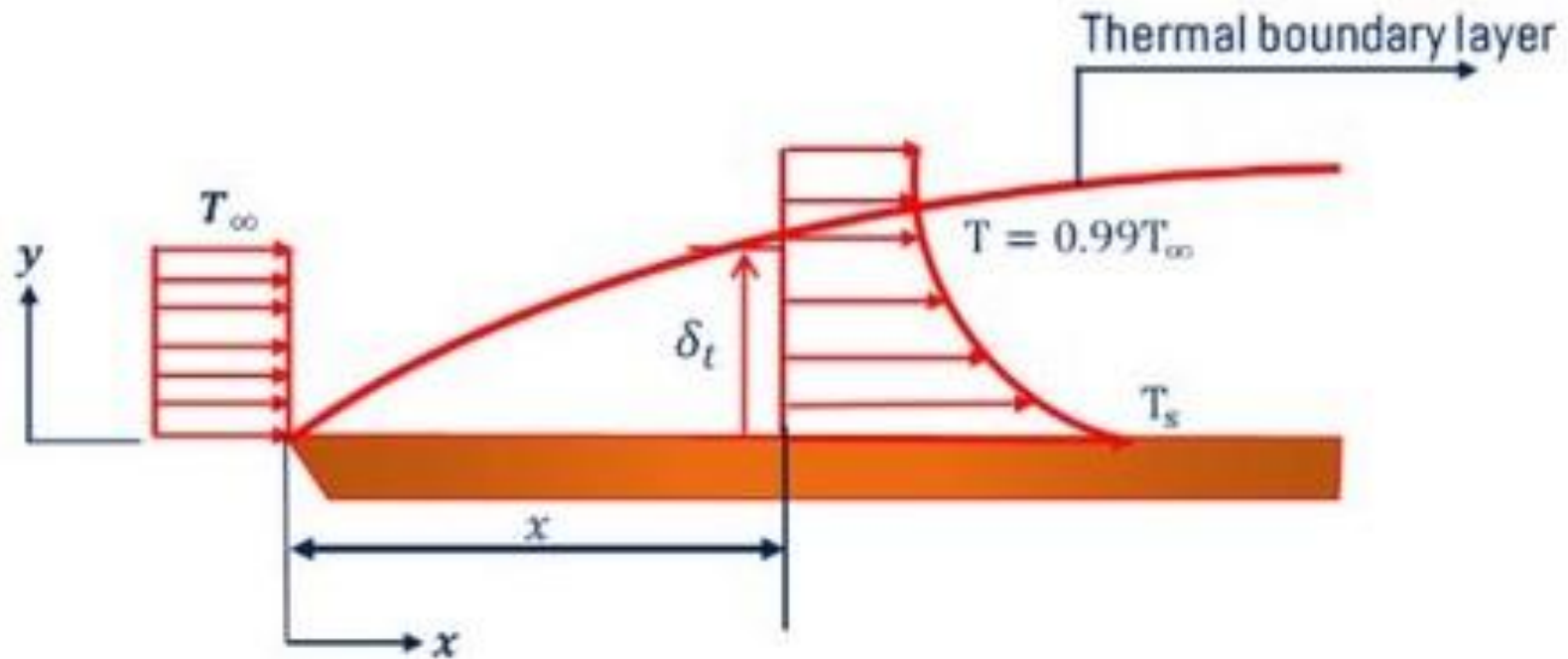
If $Re_x > Re_{cr}$, then the flow is turbulent.



Boundary Layer Thickness along the length of a Solid Surface



Thermal Boundary Layer



Pressure Drop Calculations

Darcy - Weisbach equation,

$$\frac{dp}{dx} = \frac{f}{D} \frac{\rho v^2}{2g}$$

ρ - density of the fluid (kg/m³)

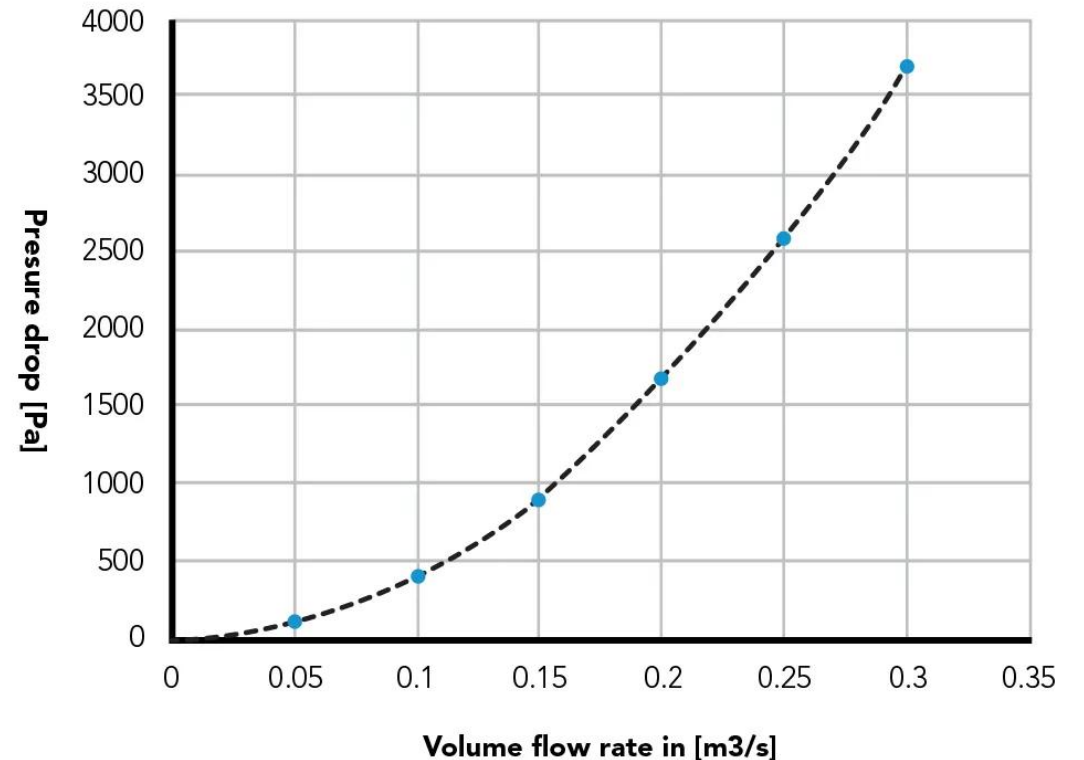
V - average velocity of the fluid (m/s)

g - acceleration due to gravity (m/s²)

f - Darcy friction factor

$f = 64/\text{Re}_D$ for laminar flow ($\text{Re}_D < 2000$)

if flow is turbulent ($\text{Re}_D > 2000$) Moody charts or
Computational Fluid Dynaics should be used to determine f



RADIATION

$$Q = \epsilon \sigma A T_s^4$$

A = Surface area of the radiating surface

σ = Stefan-Boltzmann constant (1.38×10^{-23} J/K)

T_s = Surface temperature in Kelvin (K)

ϵ = Emissivity

$$Q = \epsilon \sigma A (T_s^4 - T_{sur}^4)$$

Emissivity Values.

Material & Finish	Emissivity
Aluminum - Polished	0.04
Aluminum - Extruded	0.06
Aluminum - Anodized	0.80
Copper - Polished	0.03
Copper - Machined	0.07
Copper - Oxidized	0.78
Steel - Rolled Strip	0.55
Steel - Oxidized	0.78
Stainless Steel - A316	0.28
Nickel Plate - Dull	0.11
Tin - Bright Plate	0.04
Paint - Gloss Finish	0.89
Paint - Flat Finish	0.94

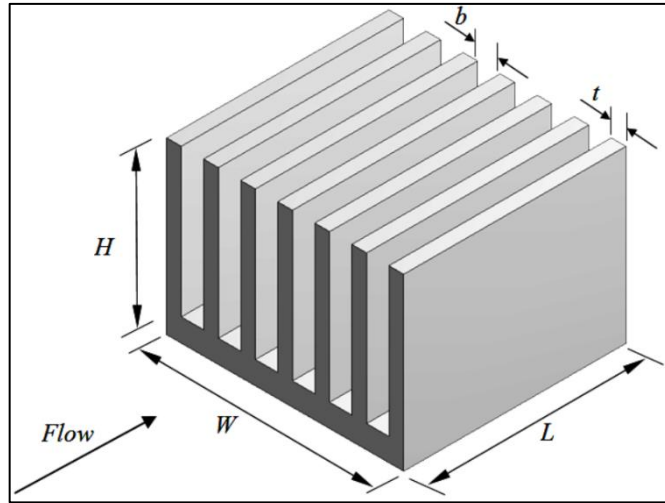
HEAT SINK DESIGN

➤ Various types of fin profiles.

➤ Factors to consider

- Fin Geometry
- Fin Density
- Fin Height
- Fin Material
- Base Plate

- Lets consider a plate finned heat sink

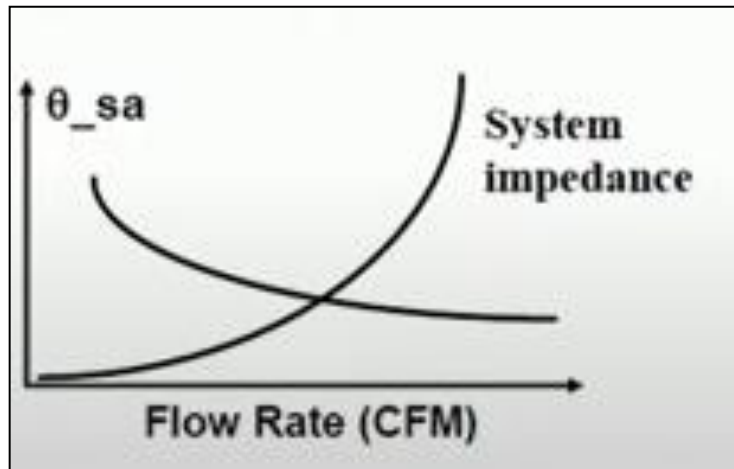


$$\theta_{sa} = \frac{T_s - T_a}{Q} = f(CFM)$$

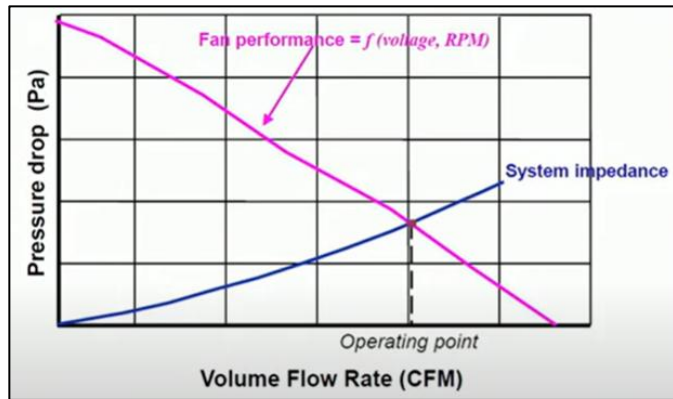
T_s - Heat sink base temperature

T_a - Approaching air temperature

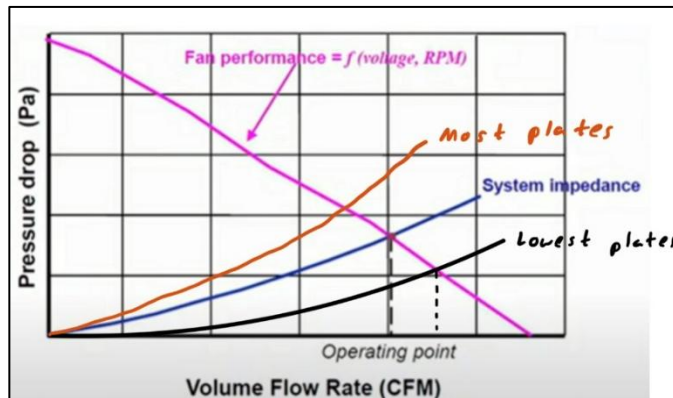
Q - Power supplied to the heat sink



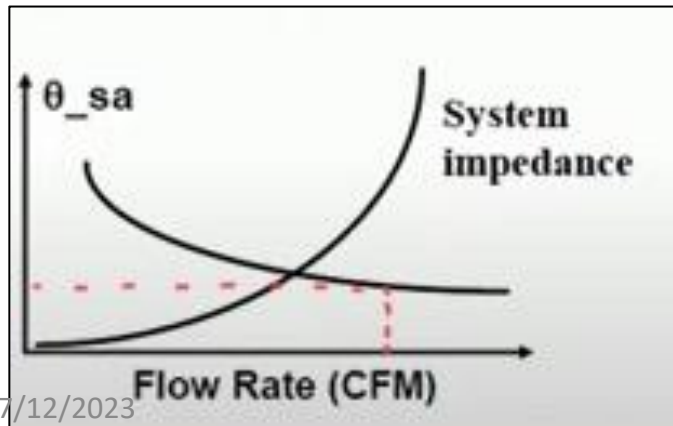
Variation of thermal resistance and the system impedance w.r.t. the flow rate



- Pressure drop is a function of fin height, width, length and fluid properties such as viscosity and density
- Intersection point is the operating point



- Pressure drop is higher when the number of fins are higher



- Corresponding flow rate value at the operating point is used to get the thermal resistance value of the heat sink

MODIFICATIONS

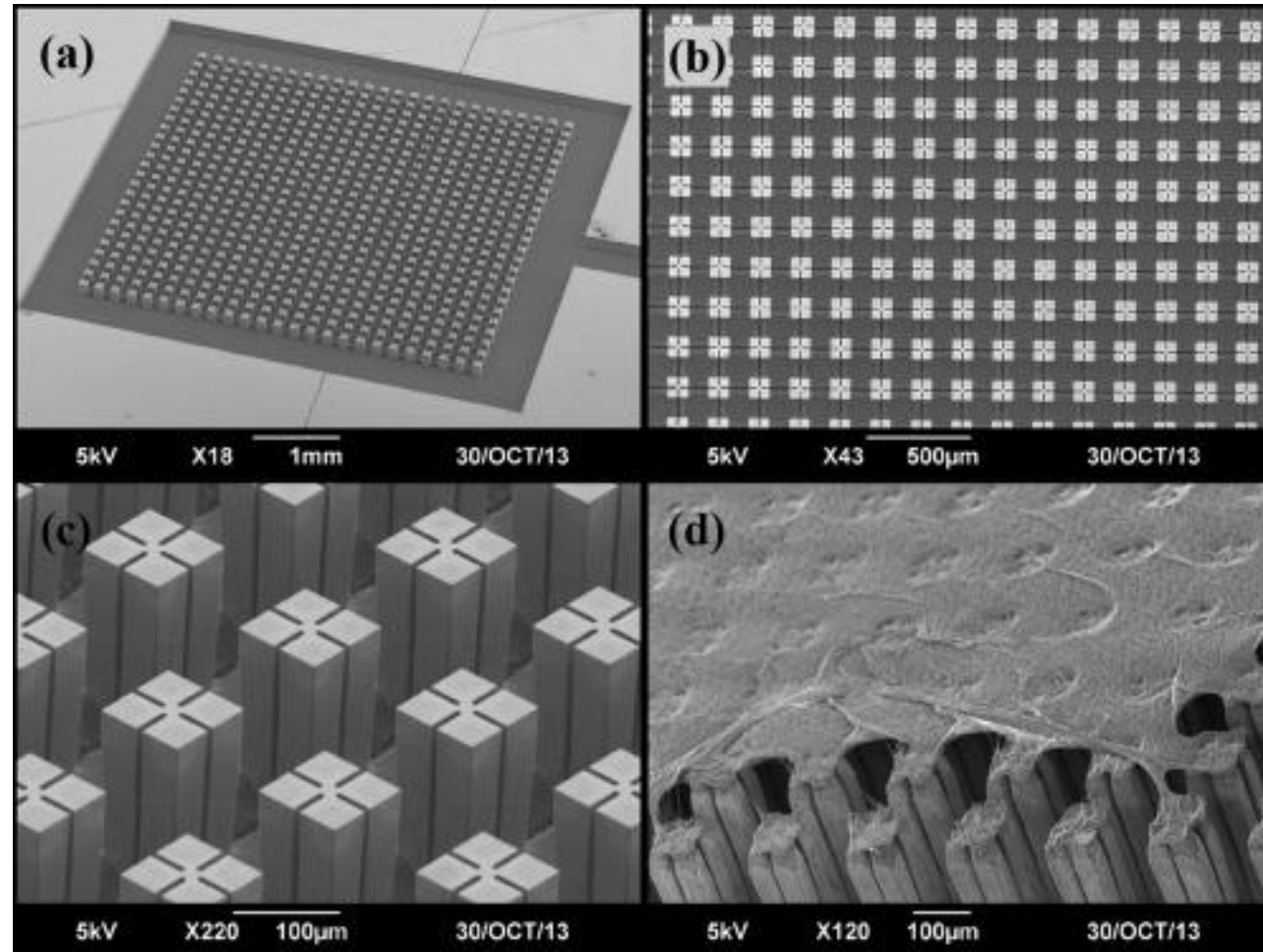
- To enhance the performance
 - Nanostructured heat sink
 - Microfluidic heat sink

- To utilize dissipated heat energy
 - Phase change materials in heat sink
 - Thermoelectric heat sinks

01) Nanostructured heat sink

- Integration of nanoscale structures on to the surface of conventional heat sinks
- Nanostructures provide more contact points for conduction between the source and the heat sink surface
- Turbulence due to the presence of nanostructures improve heat transfer through convection
- Provide larger surface area for heat transfer

Nanostructured heat sink

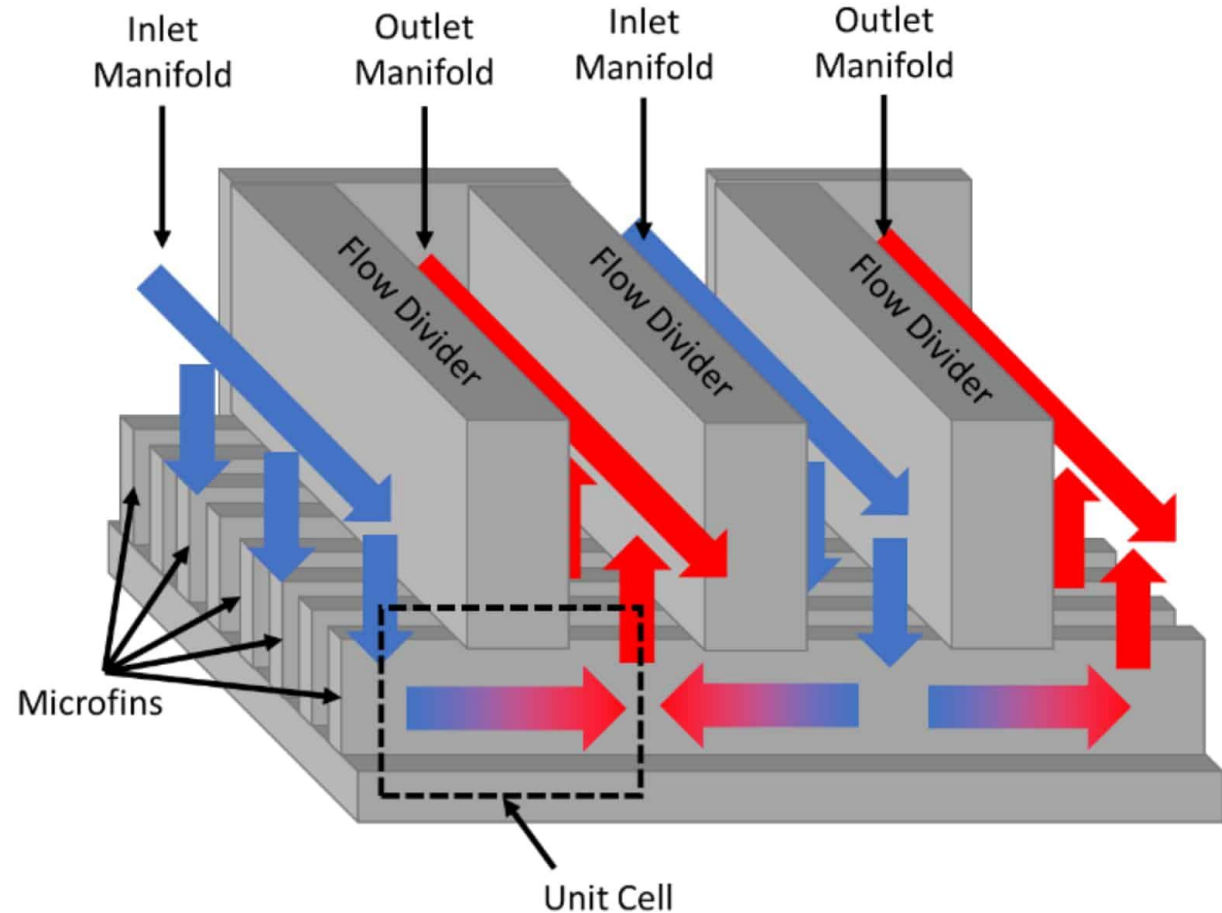


02) Microfluidic heat sink

- Utilize microchannels to enhance cooling efficiency
- Heat is transferred from the source to the fluid within the microchannels through conduction
- Convective heat transfer occurs within the fluid
- Heated fluid is directed towards a heat exchanger
- Cooled fluid repeat the heat dissipation cycle

Advantages

- ✓ Enhanced heat transfer
- ✓ Uniform cooling
- ✓ Compact size
- ✓ Flow control



Microfluidic heat sink

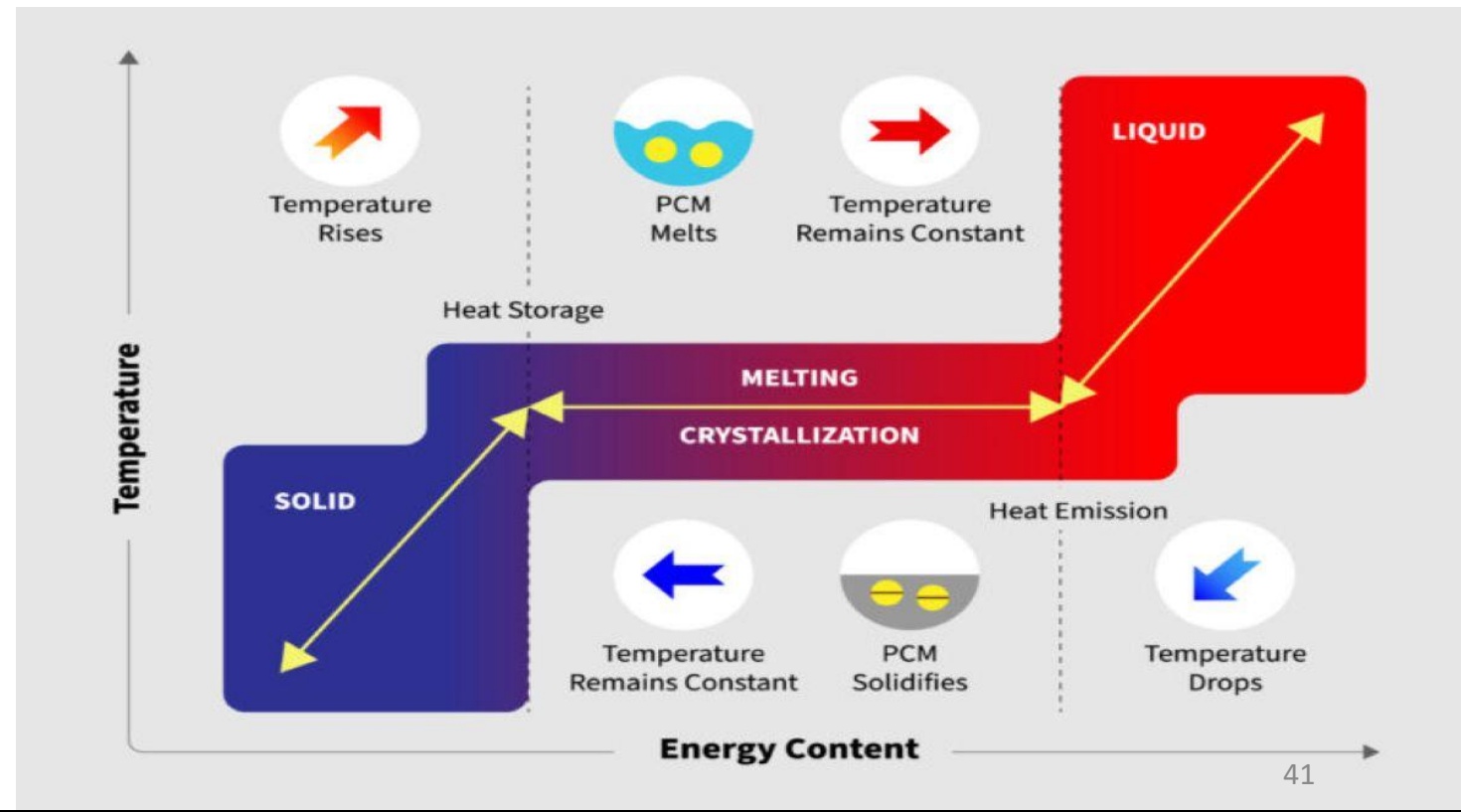
03) Phase change materials in heat sink

- PCMs can absorb or release heat during a phase transition
- Absorbed energy can be stored and dissipated over an extended period
- During the phase change, PCM absorbs heat while maintaining a nearly constant temperature (Latent heat)
- Act as thermal buffers helping to regulate temperature fluctuations and prevent rapid temperature spikes

Advantages

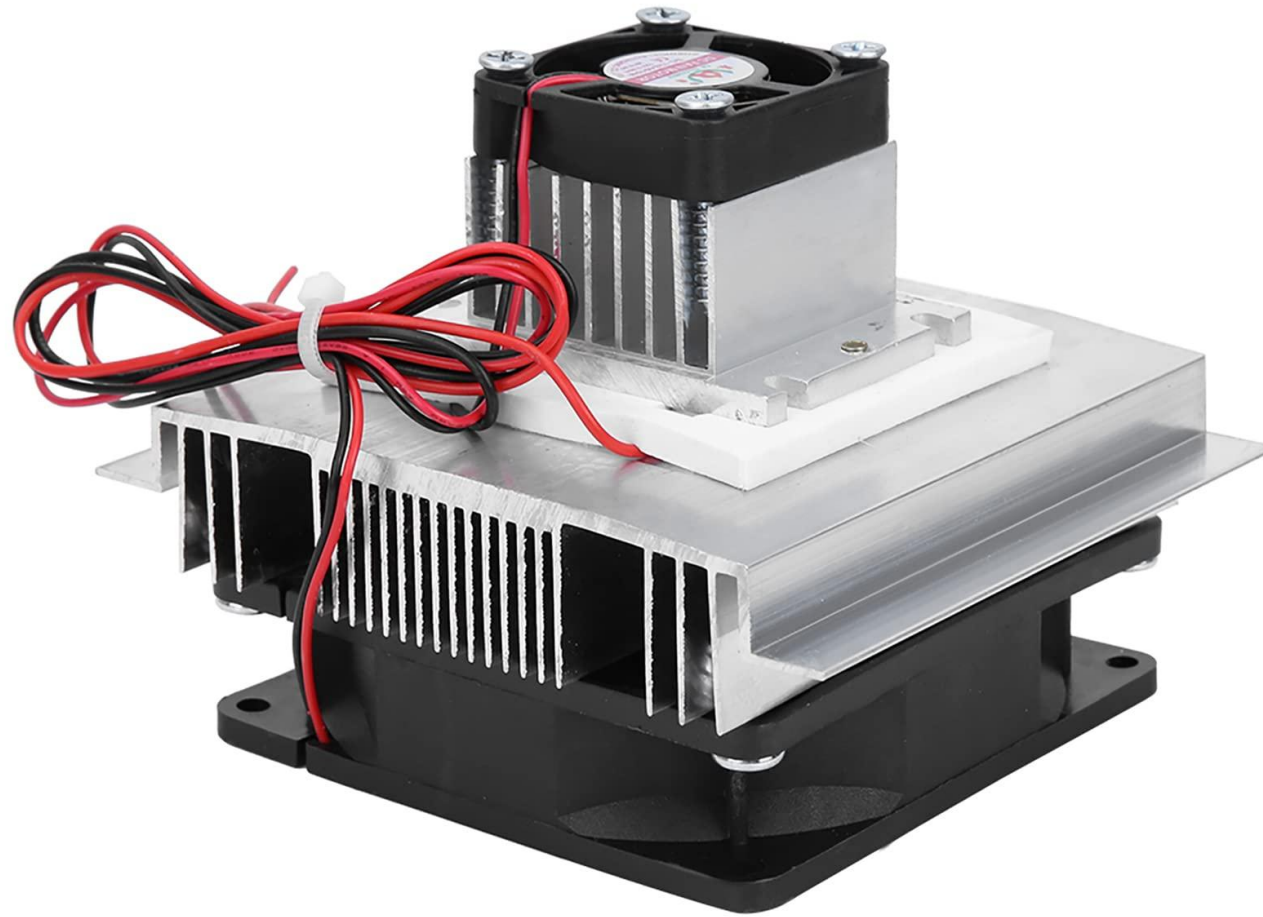
- ✓ Reduced temperature fluctuations
- ✓ Increased cooling capacity
- ✓ Thermal cycling tolerance
- ✓ Compact design

PCMs Phase transition



04) Thermoelectric heat sink

- Converts the temperature gradient into usable electrical energy through thermoelectric effect
- Thermoelectric materials can generate an electric current when exposed to a temperature gradient
- Hot side of the thermoelectric module absorbs heat from the source
- Temperature gradient is created between hot side of the module and the cold side producing a corresponding electric current
- Generated current can be used to power external devices or can be stored



Thermoelectric heat sink

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CONCLUSION

END